

25/8/25

Module - 4

Recurrence relations

$$f_n = f_{n-1} + f_{n-2}$$

$$f_0 = 0$$

$$f_1 = 1$$

$$n \geq 0$$

$$n \geq 1$$

$\Rightarrow$ Fibonacci series

2nd order, linear homogeneous.

If the value of a relation depends upon the precedence value then it is a recurrence relation.

Eg:- 5, 15, 45, ...

$a_0 = 5$ GP $\rightarrow$ geometric progression

$$r = 15/5 = 3$$

$$a_{n+1} = (r)^{n+1} a_0$$

$$a_1 = 3 \times 5$$

$$a_2 = 3 \times 3 \times 5$$

$$a_{n+1} = 3a_n$$

$\rightarrow$ In recurrence equation there is,

• homogeneous, linear, order.

homogeneous

Non-homogeneous

$$a_{n+1} = 3a_n$$

$$a_{n+1} - 3a_n = 0$$

$$f(n) = 0$$

$$a_{n+1} - 3a_n = n^2 - 1$$

$$\frac{\text{Degree}}{a_n + 2a_{n-1} + 18a_{n-2} + a_{n-3} = 0}$$

$$\text{degree} = 4$$

order.

$$\text{eg: } a_{n+1} = a_n + a_{n-2} + n$$

$\rightarrow$ highest

$\rightarrow$ lowest

highest - lowest = order

$$\text{eg: } a_{n+1} = 3a_n$$

$$n+1 - n = 1$$

$$\begin{aligned} \text{order} &= n - (n-3) \\ &= n - n + 3 \\ &= 3 \end{aligned}$$

$$n+1 - (n-2) = 3$$

Linear

$$a_{n+1} = a_n + a_{n-2} + n \rightarrow \text{Linear}$$

Non-linear

$$a_{n+1} = 3a_n a_{n-1}$$

1) First order linear recurrence relation

Homogeneous solution.

The unique solution of the recurrence relation $a_{n+1} = da_n$

where $n \geq 0$, d is a constant and $a_0 = A$ is given by

$$\{a_n = Ad^n\} \text{ for all } n \geq 0.$$

eg:- Solve the recurrence relation $a_n = 7a_{n-1}$, where $n \geq 1$

$$\text{and } a_2 = 98$$

$$\text{ans:)} \quad d = 7$$

$$a_2 = 98$$

$$a_n = 7a_{n-1}$$

$$a_2 = 7a_1 = 98$$

$$a_1 = 98/7 = \underline{\underline{14}}$$

$$\therefore a_1 = 7a_0$$

$$a_0 = 14/7 = \underline{\underline{2}}$$

$$a_0 = A = 2$$

$$d = 7$$

$$\therefore a_n = 2 \cdot 7^n, n \geq 0$$

eg:- Find the unique solution of the recurrence relation

$$6a_n - 7a_{n-1} = 0, \text{ where } n \geq 1, a_3 = 343$$

ans:)

$$a_{n+1} = da_n$$

$$d = 7/6$$

$$6a_3 = 7a_2$$

$$6a_n = 7a_{n-1}$$

$$a_3 = 343$$

$$\frac{6 \times 343}{7} = a_2 = \underline{\underline{294}}$$

$$a_n = 7/6 a_{n-1}$$

$$6a_2 = 7a_1$$

$$\frac{6 \times 294}{7} = a_1 = \underline{\underline{252}}$$

$$a_n = Ad^n$$

$$6a_1 = 7a_0$$

$$\frac{6 \times 252}{7} = a_0 = \underline{\underline{216}} = A$$

$$\therefore a_n = \frac{7}{6} \left(\frac{216}{7} \right) \left(\frac{7}{6} \right)^n$$

$n \geq 0$

Eg:- Find recurrence relation with initial conditions for

the following sequences a) 3, 7, 11, 15, 19

ans) $a_n = 4 + a_{n-1}$ $a_{n+1} = 4 + a_n, n \geq 0$

$d=1, a_0=3$

$a_0 = 3$ (or) $a_n = a_{n-1} + 4, n \geq 1$

b) 8, $24/7$, $72/49$, $216/343$

ans:) $a_{n+1} = \frac{a_n \times 3}{7}, n \geq 0$

$d=3/7, a_0=8$

Q) A person invests Rs 100,000 at 12% interest compounded annually. a) Find the amount at the end of 1st, 2nd, 3rd year. b) Write the general explicit formula. c) How long will it take to double the investment.

ans:) Let a_n represent the amount at the end of n years.

Then $a_n = a_{n-1} + (0.12)a_{n-1}$, where the second term in RHS is the interest received. Thus

$a_n = (1.12)a_{n-1}$ for $n > 1$.

a) Here the initial investment is

$a_0 = \text{Rs } 100,000$. Then

$a_1 = (1.12)a_0 = (1.12)(100,000) = 112,000$

$a_2 = (1.12)a_1 = (1.12)(112,000) = 125,440$

$a_3 = (1.12)a_2 = (1.12)(125,440) = 140,492.8$

b) $2x^2 - 5x + 2 = 0$ (Equation 1)
 $2x^2 - 4x - x + 2 = 0$
 $2x(x-2) - 1(x-2) = 0$
 $(2x-1)(x-2) = 0$ (Equation 2)
 $2x-1 = 0$ or $x-2 = 0$ (Equation 3)
 $x = \frac{1}{2}$ or $x = 2$ (Equation 4)

Roots of the equation are $x = \frac{1}{2}$ and $x = 2$ (Equation 5)
 $x = \frac{1}{2}$ and $x = 2$ (Equation 6)

Let $y_1 = e^{x/2}$ and $y_2 = e^{2x}$ (Equation 7)
 $(e^{x/2})' + (e^{2x})' = 0$ (Equation 8)

$(e^{x/2})' + (e^{2x})' = 0$ (Equation 9)

$e^{x/2} + 2e^{2x} = 0$ (Equation 10)

26/3/25

• Second order linear homogeneous recurrence relation
with constant coefficient

1) Write the characteristic polynomial.

2) Factor the polynomial and find its roots.

3) Determine the form of a_n .

Case 1: Real & distinct roots ($r_1 \neq r_2$)

Case 2: Double (or triple) roots ($r_1 = r_2$)

Case 3: Complex roots ($r_1 \neq r_2$)

4) Solve for the coefficients.

1) $a_n = 5a_{n-1} - 6a_{n-2}$ and the boundary conditions are $a_0 = 1$ & $a_1 = 1$, solve.

ans.) $x^n = 5x^{n-1} - 6x^{n-2}$

step 1) $x^n - 5x^{n-1} + 6x^{n-2} = 0$
 $\div$ whole equation by x^{n-2}

$\Rightarrow x^2 - 5x + 6 = 0$

step 2) by solving $x^2 - 5x + 6 = 0$ using quadratic equation.
 we get $x = 2, 3$

step 3) case 1:- Distinct & real roots

$$a_n = A(x_1)^n + B(x_2)^n$$

step 4) $a_n = A(2)^n + B(3)^n$

$a_0 = 1 \Rightarrow A(2)^0 + B(3)^0 = 1$

$A + B = 1$ — (1)

$a_1 = 1 \Rightarrow A(2)^1 + B(3)^1 = 1$

$2A + 3B = 1$ — (2)

$\therefore A = 2, B = -1$

$\therefore a_n = 2 \times 2^n + (-1)3^n$

$a_n = 2^{n+1} - 3^n$

By solving

① and ②

$A + B = 1$ — (1)

$2A + 3B = 1$ — (2)

① $\times 2$

$2A + 2B = 2$

$- 2A + 3B = 1$

$-B = 1$

$B = -1$

$A + (-1) = 1$

$A = 2$

2) $a_n = a_{n-1} + 6a_{n-2}$, where $n \geq 3$, $a_1 = 8$, $a_0 = 1$. solve.

ans:) $x^n = x^{n-1} + 6x^{n-2}$

step 1) $x^n - x^{n-1} - 6x^{n-2} = 0$

$\div$ whole equation by x^{n-2}

$$x^2 - x - 6 = 0$$

step 2) By solving $x^2 - x - 6 = 0$ using quadratic equation we get $x = 3, -2$

step 3) case 1 :- Distinct & real roots,

$$a_n = A(3)^n + B(-2)^n$$

Step 4) $a_0 = 1 \Rightarrow A(3)^0 + B(-2)^0 = 1$

$$A + B = 1 \text{ --- (1)}$$

$$a_1 = 8 \Rightarrow A(3)^1 + B(-2)^1 = 8$$

$$3A - 2B = 8 \text{ --- (2)}$$

By solving (1) & (2)

$$A + B = 1 \text{ --- (1) } \times 2$$

$$3A - 2B = 8 \text{ --- (2)}$$

$$2A + 2B = 2$$

$$3A - 2B = 8$$

$$5A = 10$$

$$A = 10/5 = 2$$

Substituting $A = 2$ in (2)

$$3A - 2B = 8$$

$$3(2) - 2B = 8$$

$$6 - 2B = 8$$

$$-2B = 2$$

$$B = -1$$

$$a_n = 2(3)^n + (-1)(-2)^n$$

$$2(3)^n - 2^n$$

Case 2.

1) $a_n = 6a_{n-1} - 9a_{n-2}$, where $a_0 = 1$ and $a_1 = 2$, solve.

ans:) $x^n = 6x^{n-1} - 9x^{n-2}$

Step 1) $x^n - 6x^{n-1} + 9x^{n-2} = 0$

$\div$ whole equation by x^{n-2}

$$\Rightarrow \underline{x^2 - 6x + 9 = 0}$$

Step 2) By solving $x^2 - 6x + 9 = 0$ using quadratic equation
 $x = 3, 3$

Step 3) Case 2 :- Double roots, ($r_1 = r_2$)

$$\boxed{a_n = A(x^n) + B \cdot n \cdot (x^n)}$$

Step 4) $a_n = A(3)^n + B \cdot n \cdot (3)^n$

$$a_0 = A(3)^0 + B \cdot 0 \cdot (3)^0$$

$$= A = \underline{1}$$

$$a_1 = A(3)^1 + B \cdot 1 \cdot (3)^1 = 2$$

$$= (1)(3) + B(3) = 2$$

$$3 + 3B = 2$$

$$3B = -1$$

$$B = \underline{-\frac{1}{3}}$$

$$\therefore a_n = 1(3)^n - \frac{1}{3} \cdot n \cdot (3)^n$$

$$= 3^n - \frac{3 \cdot n}{3}$$

$$= 3^n - n \cdot 3^{n-1} = \underline{\underline{3^{n-1}(3-n)}}$$

Q) $a_n = 4a_{n-1} - 4a_{n-2}$, where $n \geq 2$, $a_0 = 1$, $a_1 = 3$, solve.

Ans) step 1) $x^n = 4x^{n-1} - 4x^{n-2}$
 $x^n - 4x^{n-1} + 4x^{n-2} = 0$

$\div$ whole equation by x^{n-2}

$\Rightarrow x^2 - 4x + 4 = 0$

step 2) By solving $x^2 - 4x + 4 = 0$
 $x = 2, 2$

step 3) $a_n = A(2)^n + B \cdot n \cdot (2)^n$

$a_0 = A(2)^0 + B \cdot 0 \cdot (2)^0$

$= A = 1$

$a_1 = A(2)^1 + B \cdot 1 \cdot (2)^1$

$= 2A + 2B$

$= 2 + 2B = 3$

$2B = 1$

$B = \frac{1}{2}$

$\therefore a_n = 2^n + \frac{1}{2} \times n \times 2^n$
 $= 2^n + \frac{n \times 2^n}{2}$
 $= \frac{2^n + n \times 2^{n-1}}{1}$
 $= 2^{n-1} (2 + n)$

Q) $2a_{n+3} = a_{n+2} + 2a_{n+1} - a_n$, $n \geq 0$, $a_0 = 0$, $a_1 = 1$, $a_2 = 2$

Ans) step 1) $2x^{n+3} = x^{n+2} + 2x^{n+1} - x^n$

$2x^{n+3} - x^{n+2} - 2x^{n+1} + x^n = 0$

$\div$ whole equation by x^n

$\Rightarrow 2x^3 - x^2 - 2x + 1 = 0$

step 2) By solving $2x^3 - x^2 - 2x + 1 = 0$

we get $x = -1, 1, \frac{1}{2}$.

Step 3) case 1:

$$a_n = A(x_1)^n + B(x_2)^n + C(x_3)^n$$

Step 4) $a_n = A(-1)^n + B(1)^n + C(\frac{1}{2})^n$

$$a_0 = A(-1)^0 + B(1)^0 + C(\frac{1}{2})^0$$

$$= A + B + C = 0 \text{ --- (1)}$$

$$a_1 = A(-1)^1 + B(1)^1 + C(\frac{1}{2})^1$$

27/3/25

Case-3.

$$a) a_n = 2(a_{n-1} - a_{n-2}) \quad \forall n \geq 2, \quad a_0 = 1, a_1 = 2$$

ans:) Step 1

$$r^n = 2r^{n-1} - 2r^{n-2}$$

$$r^n - 2r^{n-1} + 2r^{n-2} = 0$$

$$\div \text{ by } r^{n-2}$$

$$r^2 - 2r + 2 = 0$$

step 2.) By solving $r^2 - 2r + 2 = 0$, using quadratic equation.
 $r = 1+i, 1-i$

step 3) case 3 :- complex roots

$$r_1 = p+iq$$

$$r_2 = p-iq$$

$$a_n = r^n (A \cos n\theta + B \sin n\theta)$$

$$\text{where } r = \sqrt{p^2 + q^2}$$

$$\theta = \tan^{-1}(q/p)$$

step 4) $r = \sqrt{p^2 + q^2}$

$$= \sqrt{1^2 + 1^2} = \underline{\underline{\sqrt{2}}}$$

$$\theta = \tan^{-1}(1/1) = \tan^{-1}(1) = \underline{\underline{45^\circ}}$$

$$a_n = \sqrt{2}^n (A \cos n 45 + B \sin n 45)$$

$$a_0 = \sqrt{2}^0 (A \cos(0) 45 + B \sin(0) 45)$$

$$= 1 (A \cos 0 + B \sin 0)$$

$$= 1 (A + 0)$$

$$= \underline{\underline{A = 1}}$$

$$a_1 = \sqrt{2}^1 (A \cos(1)(45) + B \sin(1)(45))$$

$$= \sqrt{2} \left(A \times \frac{\sqrt{2}}{2} + B \times \frac{\sqrt{2}}{2} \right)$$

$$= \sqrt{2} \times \frac{\sqrt{2}}{2} (A + B)$$

$$A + B = 2$$

$$\Rightarrow \underline{\underline{B = 1}}$$

$$\therefore a_n = \sqrt{2}^n (\cos n45 + \sin n45)$$

• Non-homogeneous recurrence relation

$$1) a_{n+2} - 5a_{n+1} + 6a_n = 2, \quad a_0 = 1, a_1 = -1$$

ans: $\rightarrow$ Second order

$\rightarrow$ linear.

$\rightarrow$ Non-homogeneous

$$a_n = a_n^h + a_n^p$$

$$\underline{a_n^h}$$

$$a_{n+2} - 5a_{n+1} + 6a_n = 0$$

$$\underline{a_n^p} \quad 2.$$

$$\underline{a_n^h}$$

$$a_{n+2} - 5a_{n+1} + 6a_n = 0$$

$$\text{Step 1) } x^{n+2} - 5x^{n+1} + 6x^n = 0$$

$$\div x^n$$

$$x^2 - 5x + 6 = 0$$

Step 2) By solving we get $x = 2, 3$.

Step 3) Case 1

$$a_n = A(x_1)^n + B(x_2)^n$$

$$= A(2)^n + B(3)^n$$

$$a_n = a_n^h + a_n^p$$

Finding solution for $a_n = A(2)^n + B(3)^n + 1$

$$\underline{a_0}$$

$$a_0 = A(2)^0 + B(3)^0 + 1$$

$$= A + B + 1 = 1$$

$$A + B = 0 \quad \text{--- (1)}$$

$$\underline{a_1}$$

$$a_1 = A(2)^1 + B(3)^1 + 1$$

$$= 2A + 3B + 1 = -1$$

$$2A + 3B = -2 \quad \text{--- (2)}$$

$$\underline{a_n^p}$$

$f(n) = \text{constant}$

$$f(n) = 2$$

constant

$$a_n = A_0$$

Applying A_0 in the relation

$$a_{n+2} - 5a_{n+1} + 6a_n = 2$$

$$\Rightarrow A_0 - 5A_0 + 6A_0 = 2$$

$$\Rightarrow 2A_0 = 2$$

$$A_0 = 1$$

Solving ① and ②

$$A + B = 0 \quad \times 2$$

$$2A + 3B = -2$$

$$\begin{array}{r} 2A + 2B = 0 \\ -2A + 3B = -2 \\ \hline -B = -2 \end{array}$$

$$-B = -2$$

$$B = \underline{\underline{-2}}$$

$$2A + 2(-2) = 0$$

$$2A - 4 = 0$$

$$2A = 4$$

$$A = \underline{\underline{2}}$$

$$\therefore a_n = 2 \cdot (2)^n + (-2) \cdot 3^n + 1$$

$$a_n = \underline{\underline{2^{n+1} - 2 \cdot 3^n + 1}}$$

$$2) a_n + 4a_{n-1} + 4a_{n-2} = 8, \quad n \geq 2, \quad a_0 = 1, \quad a_1 = 2$$

Ans.) Step 1) $x^n + 4x^{n-1} + 4x^{n-2} = 80 \frac{a_n h.}{a_n}$

$$\frac{x^n}{x^{n-2}} + 4 \frac{x^{n-1}}{x^{n-2}} + 4 \frac{x^{n-2}}{x^{n-2}} = 80$$

$$x^2 + 4x + 4 = 80$$

$$x^2 + 4x + 4 = 0$$

$$a_n = a_n^h + a_n^p$$

$$a_n^h = a_n + 4a_{n-1} + 4a_{n-2}$$

$$a_n^p = 8$$

Step 2) By solving we get $x = -2, -2$.

Step 3) Case 2

$$a_n = A(x^n) + B \cdot n \cdot (x^n)$$

$$a_n = A(-2)^n + B \cdot n \cdot (-2)^n$$

Applying A_0 in the relation,

$$\Rightarrow A_0 + 4A_0 + 4A_0 = 8$$

$$\Rightarrow 9A_0 = 8$$

$$\Rightarrow A_0 = \underline{\underline{8/9}}$$

$$a_n = a_n^h + a_n^p$$

$$= A(-2)^n + B \cdot n \cdot (-2)^n + 8/9$$

$$\underline{a_0}$$

$$a_0 = A(-2)^0 + B \cdot 0 \cdot (-2)^0 + 8/9$$

$$= A + 8/9 = 1$$

$$A = 1 - \frac{8}{9} = \underline{\underline{1/9}}$$

$$\underline{a_1}$$

$$a_1 = A(-2)^1 + B \cdot 1 \cdot (-2)^1 + 8/9$$

$$= -2A + -2B + 8/9 = 2$$

$$= -2(1/9) - 2B + 8/9 = 2$$

$$= -\frac{2}{9} + 8/9 - 2B = 2$$

$$= \frac{6}{9} - 2B = 2$$

$$= -2B - 2\frac{6}{9} = \frac{10}{9}$$

$$B =$$

$$\frac{2}{3} - 2B = 2$$

$$\frac{2}{3} - 2 = 2B$$

$$-\frac{4}{3} = 2B$$

$$\therefore B = -4/6 = \underline{\underline{-2/3}}$$

$$\therefore a_n = \frac{1}{9}(-2)^n - \frac{2}{3}n(-2)^n$$

$$+ 8/9$$

$$3) \text{ solve } a_n = 3a_{n-1} + 2n, a_1 = 3$$

$$a_n = a_n^h + a_n^p$$

$$a_n^h = 3a_{n-1} + 2n = a_n$$

$$a_n^p = a_n$$

$$a_n^h$$

$$3a_{n-1} + 2n = 0$$

$$\frac{n-1}{2} + 2$$

$$a_n = 3a_{n-1} + 2n$$

$$2^n = 3 \cdot 2^{n-1} + 0$$

$$2^n = 3 \cdot 2^{n-1}$$

$$2^n - 3 \cdot 2^{n-1} = 0$$

$$\frac{2}{2} \cdot 2^{n-1}$$

$$2 - 3 = 0$$

$$\therefore 2 = 3$$

(case 1:) $a_n = A(3)^n$

$$a_n = a_n^h + a_n^p = A(3)^n - \frac{3}{2} - n$$

$$= A(3)^n - \frac{3}{2} - n$$

$$a_0$$

$$A(3)^0 - \frac{3}{2} - 1(0)$$

$$A_n = -\frac{3}{2} - n$$

$$A - \frac{3}{2} =$$

$$a_1$$

$$A(3)^1 - \frac{3}{2} - 1(1) = 3$$

$$3A - \frac{3}{2} - 1 = 3$$

$$3A - \left(-\frac{1}{2}\right) = 3$$

$$3A + \frac{1}{2} = 3$$

$$3A = 3 - \frac{1}{2} = \frac{5}{2}$$

$$A = \frac{5}{6}$$

$$a_n^p$$

$$f(n) = n$$

$$f(n) = 2n$$

$$a_n^p = A_0 + A_1 n$$

Applying in recurrence relation.

$$a_n = 3a_{n-1} + 2n$$

$$a_n - 3a_{n-1} = 2n$$

$$A_0 + A_1 n - 3(A_0 + A_1(n-1)) = 2n$$

$$A_0 + A_1 n - 3A_0 - 3A_1 n + 3A_1 = 2n$$

$$-2A_0 - 2A_1 n + 3A_1 = 2n$$

$$-2A_1 = 2$$

$$A_1 = -1 //$$

$$-2A_0 + 3A_1 = 0$$

$$-2A_0 + 3(-1) = 0$$

$$-2A_0 - 3 = 0$$

$$-2A_0 = 3$$

$$A_0 = -\frac{3}{2}$$

$$A_n = -\frac{3}{2} - n$$

$$A_n = A_0 + A_1 n$$

$$= -\frac{3}{2} - n$$

$$= -2 - (-3n)$$

$$= -2 + 3n$$

a1

$$A(3) - \frac{3-1}{2} = 3$$

$$3A - \frac{5}{2} = 3$$

$$3A = 3 + \frac{5}{2} = \frac{11}{2}$$

~~A =~~

$$A = \frac{11}{6}$$

$$a_n = \frac{11}{6}(3)^n - \frac{3}{2}n$$

$$4) a_n + 4a_{n-1} + 4a_{n-2} = 8n, a_0 = 1, a_1 = 2$$

$$a_n = a_n^h + a_n^p$$

$$a_n^h = a_n + 4a_{n-1} + 4a_{n-2} = 0$$

$$a_n^p = 8n$$

a_n^h

$$a_n + 4a_{n-1} + 4a_{n-2} = 0$$

$$x^n + 4x^{n-1} + 4x^{n-2} = 0$$

$$\div \text{ by } x^{n-2}$$

$$x^2 + 4x + 4 = 0$$

By solving we get $x = -2, -2$.

Case 2.

$$a_n = A(-2)^n + B \cdot n \cdot (-2)^n$$

a_n^p

$$f(n) = 8n$$

$$a_n = A_0 + A_1 n$$

Applying in recurrence relation

$$a_n + 4a_{n-1} + 4a_{n-2} = 8n$$

$$A_0 + A_1 n + 4(A_0 + A_1(n-1))$$

$$+ 4(A_0 + A_1(n-2)) = 8n$$

$$A_0 + A_1 n + 4A_0 + 4A_1 n - 4A_1$$

$$+ 4A_0 + 4A_1 n - 8A_1 = 8n$$

$$9A_0 + 9A_1n$$

$$9A_0 - 12A_1 = 0 \quad \text{--- (1)}$$

$$9A_1n = 8n$$

$$9A_1 = 8$$

$$A_1 = \underline{\underline{8/9}}$$

substituting $A_1 = 8/9$ in (1)

$$9A_0 - 12 \times \frac{8}{9} = 0$$

$$9A_0 - \frac{96}{9} = 0$$

$$9A_0 = \frac{96}{9}$$

$$A_0 = \frac{96}{81} = \underline{\underline{\frac{32}{27}}}$$

$$a_n = \underline{\underline{\frac{32}{27} + \frac{8}{9}n}}$$

$$\begin{aligned} a_n &= a_n^h + a_n^p \\ &= A(-2)^n + B \cdot n \cdot (-2)^n \\ &\quad + \frac{32}{27} + \frac{8}{9}n \end{aligned}$$

$$\underline{a_0}$$

$$\begin{aligned} a_0 &= A(-2)^0 + B \cdot 0 \cdot (-2)^0 \\ &\quad + \frac{32}{27} + \frac{8}{9}(0) \end{aligned}$$

$$= A + \frac{32}{27} = 1$$

$$A = 1 - \frac{32}{27} = \underline{\underline{-5/27}}$$

$$\underline{a_1}$$

$$\begin{aligned} a_1 &= A(-2)^1 + B \cdot (1) \cdot (-2)^1 \\ &\quad + \frac{32}{27} + \frac{8}{9}(1) \end{aligned}$$

$$= -2A - 2B + \frac{32}{27} + \frac{8}{9}$$

$$= -2(-5/27) - 2B + \frac{56}{27}$$

$$= \frac{10}{27} + \frac{56}{27} - 2B$$

$$= \frac{66}{27} - 2B = 2$$

$$\frac{66}{27} - 2 = 2B$$

$$\frac{4}{9} = 2B$$

$$\therefore B = \frac{4}{18} = \underline{\underline{2/9}}$$

$$\therefore a_n = (-5/27)(-2)^n + (2/9)n \cdot (-2)^n + \frac{32}{27} + \frac{8}{9}n$$

28/3/25 Non-homogeneous recurrence relation with $f(n) = n^2$

$$e) a_{n+2} - 10a_{n+1} + 21a_n = 3n^2 - 2$$

$$a_n = a_n^h + a_n^p$$

$$\underline{a_n^h}$$

$$a_{n+2} - 10a_{n+1} + 21a_n = 0$$

$$x^{n+2} - 10x^{n+1} + 21x^n = 0$$

$$\frac{x^n}{x^n} \cdot n$$

$$x^2 - 10x + 21 = 0$$

By solving $x = \underline{7, 3}$.

$$a_n = A(7)^n + B(3)^n$$

$$(A_0 + A_1(n+2) + A_2(n+2)^2)$$

$$= A_0 + A_1n + 2A_1 + A_2(n^2 + 4 + 4n)$$

$$A_0 + A_1n + 2A_1 + A_2n^2 + 4A_2 + 4A_2n \quad \text{--- (1)}$$

$$10[A_0 + A_1(n+1) + A_2(n+1)^2]$$

$$= 10[A_0 + A_1n + A_1 + A_2(n^2 + 1 + 2n)]$$

$$10[A_0 + A_1n + A_1 + A_2n^2 + A_2 + 2A_2n]$$

$$- [10A_0 + 10A_1n + 10A_1 + 10A_2n^2 + 10A_2 + 20A_2n] \quad \text{--- (2)}$$

$$21[A_0 + A_1n + A_2n^2]$$

$$= 21A_0 + 21A_1n + 21A_2n^2 \quad \text{--- (3)}$$

$$\underline{a_n^p}$$

$$a_n = A_0 + A_1n + A_2n^2$$

Applying in recurrence relation

$$a_{n+2} - 10a_{n+1} + 21a_n = 3n^2 - 2$$

$$A_0 + A_1(n+2) + A_2(n+2)^2 -$$

$$10[A_0 + A_1(n+1) + A_2(n+1)^2] +$$

$$21[A_0 + A_1n + A_2n^2] = 3n^2 - 2$$

$$① - ② + ③$$

$$12A_0 + 12A_1n - 8A_1 + 12A_2n^2 - 16A_2 - 16A_2n = 3n^2 - 2 \quad \text{--- (4)}$$

$$12A_2n^2 = 3n^2$$

$$12A_2 = 3$$

$$A_2 = 3/12 = \frac{1}{4}$$

$$\text{Substitute } A_2 = \frac{1}{4} \text{ in (4)}$$

$$12A_0 + 12A_1n - 8A_1 +$$

$$12A_1n - 16A_2n = 0$$

$$12A_1n - 16(\frac{1}{4})n = 0$$

$$12A_1n - 4n = 0$$

$$n(12A_1 - 4) = 0$$

$$12A_1 - 4 = 0$$

$$12A_1 = 4$$

$$A_1 = 4/12 = \frac{1}{3}$$

$$\text{Substituting } A_1 = \frac{1}{3} \text{ and } A_2 = \frac{1}{4}$$

$$12A_0 - 8A_1 - 6A_2 = -2$$

$$12A_0 - 8(\frac{1}{3}) - 6(\frac{1}{4}) = -2$$

$$12A_0 - \frac{8}{3} - \frac{3}{2} = -2$$

$$12A_0 - \frac{25}{6} = -2$$

$$12A_0 = -2 + \frac{25}{6} = \frac{13}{6}$$

$$A_0 = \frac{13}{72}$$

$$\therefore a_n = a_n^h + a_n^p$$

$$= A(7)^n + B(3)^n + A_0 + A_1n + A_2n^2$$

$$= A(7)^n + B(3)^n + \frac{13}{72} + \frac{1}{3}n + \frac{1}{4}n^2$$

C, a constant

$$n$$

$$n^2$$

$$n^t, t \in \mathbb{Z}^+$$

$$x^n, x \in \mathbb{R}$$

$$\sin \theta n$$

$$\cos \theta n$$

$$n^t x^n$$

$$x^n \sin \theta n$$

$$x^n \cos \theta n$$

$$a_n^p$$

A, a constant

$$A_1 n + A_0$$

$$A_2 n^2 + A_1 n + A_0$$

$$A_t n^t + A_{t-1} n^{t-1} + \dots + A_1 n + A_0$$

$$A x^n$$

$$A \sin \theta n + B \cos \theta n$$

$$A \sin \theta n + B \cos \theta n$$

$$x^n (A_t n^t + A_{t-1} n^{t-1} + \dots + A_1 n + A_0)$$

$$A x^n \sin \theta n + B x^n \cos \theta n$$

$$A x^n \sin \theta n + B x^n \cos \theta n$$

2/4/25 $\circ$ solve $a_n = 2a_{n-1} + 2^n$

ans: $a_n = a_n^h + a_n^p$

$$\underline{a_n^h}$$

$$a_n = 2a_{n-1}$$

$$a_n^h = 2^n A$$

$$\underline{a_n^p}$$

$$a_n = A_0 \cdot n \cdot 2^n$$

Applying in recurrence relation

$$A_0 \cdot n \cdot 2^n = 2 \cdot A_0 (n-1) 2^{n-1} + 2^n$$

$$A_0 \cdot n \cdot 2^n = 2$$

11/4/25 Generating functions

Let $a_0, a_1, a_2, \dots$ be a sequence of real numbers,

then the generating function of the sequence is defined as

$$g(x) = a_0 + a_1x + a_2x^2 + \dots = \sum_{i=0}^{\infty} a_i x^i$$

{Note:- Expansion of $\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots$ }

① Generative function of the sequence $1, 1, 1, 1, \dots$
 $= g(x) = \frac{1}{1-x}$

② $\frac{d}{dx} \frac{1}{1-x} = \frac{d}{dx} [1 + x + x^2 + \dots]$ G.f of the sequence $1, 2, 3, 4$
 $\frac{-1}{(1-x)^2} = 0 + 1 + 2x + 3x^2 + \dots$
 $\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \dots$
 $g(x) = \frac{1}{(1-x)^2}$

③ $\frac{x}{(1-x)^2} = x \left[\frac{1}{(1-x)^2} \right]$ is the G.f of the sequence $0, 1, 2, 3, 4, \dots$
 $= x [1 + 2x + 3x^2 + 4x^3 + \dots]$
 $= x + 2x^2 + 3x^3 + 4x^4 + \dots$

④ $\frac{d}{dx} \frac{x}{(1-x)^2} = \frac{d}{dx} (x + 2x^2 + 3x^3 + \dots)$ G.f of the sequence $1^2, 2^2, 3^2, \dots$
 $\frac{1+x}{(1-x)^3} = 1 + 2^2x + 3^2x^2 + \dots$
 $g(x) = \frac{1+x}{(1-x)^3}$

⑤ $\frac{x(1+x)}{(1-x)^3} = x \left[1 + 2^2x + 3^2x^2 + \dots \right]$
 $= x + 2^2x^2 + 3^2x^3 + \dots$

$\frac{x(1+x)}{(1-x)^3}$ is the G.f of the sequence $0, 1^2, 2^2, 3^2, 4^2, \dots$

$$\left\{ \text{Note:- } \frac{1}{1+x} = 1 - x + x^2 - x^3 + x^4 - x^5 + \dots \right\}$$

① G.f of the sequence $1, -1, 1, -1, 1, -1, \dots$ $g(x) = \frac{1}{1+x}$.

$$\rightarrow (1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots$$

$(1+x)^n$ is the g.f of the sequence ${}^nC_0, {}^nC_1, {}^nC_2, \dots$

$\rightarrow$ Find the g.f of the sequence $0, 0, 0, 1, 1, 1, 1, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 0 + 0x + 0x^2 + 1x^3 + 1x^4 + 1x^5 + \dots \\ &= x^3 + x^4 + x^5 + \dots \\ &= x^3 [1 + x + x^2 + \dots] \\ &= x^3 \frac{1}{1-x} \\ &= \frac{x^3}{1-x} \end{aligned}$$

$\rightarrow$ Find the g.f of the sequence $1, 1, 1, 3, 1, 1, 1, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 1 + 1x + 1x^2 + 3x^3 + 1x^4 + 1x^5 + \dots \\ &= 1 + x + x^2 + 3x^3 + x^4 + x^5 + \dots \\ &= (1 + x + x^2 + x^3 + x^4 + x^5 + \dots) + 2x^3 \\ &= \left(\frac{1}{1-x} \right) + 2x^3 \\ &= \frac{1 + 2x^3(1-x)}{1-x} \\ &= \frac{1 + 2x^3 - 2x^4}{1-x} = \frac{2x^2(1+2x-2x^2)}{1-x} \end{aligned}$$

Exponential generating function

Let $a_0, a_1, a_2, a_3, \dots$ be a real numbers then

$$g(x) = a_0 + a_1 \frac{x}{1!} + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \dots$$

is called exponential generating function of the given sequence.

→ Find e.g.f of the sequence $1, 1, 1, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \\ &= e^x. \end{aligned}$$

→ Find e.g.f of the sequence $1, -1, 1, -1, 1, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 1 + (-1) \frac{x}{1!} + (+1) \frac{x^2}{2!} + (-1) \frac{x^3}{3!} + 1 \frac{x^4}{4!} + \dots \\ &= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} = e^{-x}. \end{aligned}$$

→ Find e.g.f of the sequence $1, 2, 2^2, 2^3, 2^4, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 1 + 2 \frac{x}{1!} + 2^2 \frac{x^2}{2!} + 2^3 \frac{x^3}{3!} + \dots \\ &= 1 + \frac{2x}{1!} + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \dots \\ &= e^{2x}. \end{aligned}$$

→ Find e.g.f of the sequence $1, -a, a^2, -a^3, a^4, \dots$

$$\begin{aligned} \text{ans:)} \quad g(x) &= 1 + (-a) \frac{x}{1!} + a^2 \frac{x^2}{2!} + (-a)^3 \frac{x^3}{3!} + \dots \\ &= 1 - \frac{ax}{1!} + \frac{(a^2x)^2}{2!} - \frac{(ax)^3}{3!} + \dots \\ &= e^{-ax}. \end{aligned}$$

→ $1, a^2, a^4, a^6, \dots$ Find e.g.f.

$$\begin{aligned} \text{ans)} \quad g(x) &= 1 + \frac{a^2(x)}{1!} + \frac{a^4(x^2)}{2!} + \frac{a^6(x^3)}{3!} + \dots \\ &= 1 + \frac{a^2 x}{1!} + \dots \\ &= e^{a^2 x} \end{aligned}$$

→ $a, a^3, a^5, a^7, \dots$ Find e.g.f.

$$\begin{aligned} \text{ans)} \quad g(x) &= a + \frac{a^3(x)}{1!} + \frac{a^5(x^2)}{2!} + \frac{a^7(x^3)}{3!} + \dots \\ &= a \left[1 + \frac{a^2(x)}{1!} + \frac{a^4(x^2)}{2!} + \frac{a^6(x^3)}{3!} + \dots \right] \\ &= a e^{a^2 x} \end{aligned}$$

→ $0, 1, 2(2), 3(2^2), 4(2^3), \dots$ Find e.g.f.

$$\begin{aligned} \text{ans)} \quad g(x) &= 0 + \frac{1(x)}{1!} + \frac{2(2)(x^2)}{2!} + \frac{3(2^2)(x^3)}{3!} + \dots \\ &= x \cdot e^{2x} \end{aligned}$$

15/4/25

Q) Determine the sequence generated by the following generating function.

a) $f(x) = 5e^{5x}$

b) $f(x) = 7e^{8x} - 4e^{3x}$

c) $f(x) = e^{3x} - 28x^3 - 6x^2 + 9x$

$$\begin{aligned} \text{ans)} \quad a) \quad 5e^{5x} &= 5 \left[1 + \frac{5x}{1!} + \frac{(5x)^2}{2!} + \frac{(5x)^3}{3!} + \dots \right] \\ &= 5 + \frac{5^2 x}{1!} + \frac{5^3 x^2}{2!} + \dots \end{aligned}$$

Sequence = $5, 5^2, 5^3, 5^4, \dots$

$$\begin{aligned}
 b) f(x) &= 7e^{8x} - 4e^{3x} \\
 &= 7 \left(1 + \frac{8x}{1!} + \frac{(8x)^2}{2!} + \dots \right) - 4 \left(1 + \frac{3x}{1!} + \frac{(3x)^2}{2!} + \dots \right) \\
 &= 7 + 7 \cdot 8 \cdot \frac{x}{1!} + 7 \cdot 8^2 \cdot \frac{x^2}{2!} + \dots - \left(4 + 4 \cdot 3 \cdot \frac{x}{1!} + 4 \cdot 3^2 \cdot \frac{x^2}{2!} + \dots \right) \\
 &= 3 + (7(8) - 4(3)) \frac{x}{1!} + (7 \cdot 8^2 - 4 \cdot 3^2) \frac{x^2}{2!} + \dots \\
 &= 3, 44, 412, \dots \rightarrow \text{sequence.}
 \end{aligned}$$

$$\begin{aligned}
 c) f(x) &= e^{3x} - 28x^3 - 6x^2 + 9x \\
 &= 1 + \frac{3x}{1!} + \frac{(3x)^2}{2!} + \dots - 28x^3 - 6x^2 + 9x \\
 &= 1 + \left(\frac{3x}{1!} + 9x \right) + \left(\frac{(3x)^2}{2!} - 6x^2 \right) + \left(\frac{(3x)^3}{3!} - 28x^3 \right) + \dots \\
 &= 1, 12, -3, -144, 3^4, 3^5, \dots
 \end{aligned}$$

2) Find the sequence generated by the exponential generating function.

ans) $1 + x + x^2 + x^3 + \dots$